nilenso

a quick showcase

About us

We are a boutique technology consultancy founded in 2013. We are known for being a trusted team of highly capable individuals that can come in, adapt quickly, and build mission-critical software that scales, and lasts. We have worked with well known companies in the US (bay area), India, and south-east Asia.

gojek
Paytm insider
Staples
grofers
PayU
PORTER
Vital Strategies
exotel
JUSPAY
clarisights
DUNZO
PROACTIVE for her
mapbox
CONSOLIDATE HEALTH
Flipkart
ASHOKA
mnyway
Milaap
Quintype
Akvo.

A tailored deck

Rather than a holistic overview of our repertoire, we present here a focused list of clients and case studies for more appropriate details.

We have deep expertise in healthcare technology
Having spent 7 years as the core engineering team behind Simple.org, an award-winning platform serving 6+ million patients across 4 countries, funded by Bloomberg, Gates Foundation, and Chan Zuckerberg Initiative. Our work has been published in Nature's Journal of Human Hypertension.

We understand US healthcare requirements
We have been working with Consolidate Health since 2023, a US-based healthtech company, building HIPAA-compliant, FHIR R4-compliant systems for EHR data consolidation, including AI-powered patient summarization.

We have proven experience with large US enterprises
Our 5-year engagement with Staples involved building experimentation platforms, ML-powered recommendation engines, and data warehousing systems at scale.

Simple

Award-Winning Global Healthcare Platform

Simple | Overview

Simple is an open-source, award-winning healthcare platform built for Resolve to Save Lives to prevent heart attacks and strokes through hypertension control. Funded by Bloomberg, Gates Foundation, and Chan Zuckerberg Initiative, the app enables healthcare workers in resource-constrained settings to manage blood pressure measurements and medications.

We have been the core engineering team on Simple.org since 2018, a 7-year engagement with a globally-funded public health initiative.

Recognition & Awards

- Peer-reviewed research published in Nature journal, Journal of Human Hypertension, 2024
- Recognized as a Digital Public Good by the UN-endorsed Digital Public Goods Alliance
- Top Prize, Interaction 20 Conference (Milan) from the global Interaction Design Association

Simple | Scale and Impact

Metric	Achievement
Patients Registered	6+ million across India, Bangladesh, Ethiopia, and Sri Lanka
Health Facilities	5,000+ public health facilities actively using the platform
Registration Time	50 seconds for new patients, 14 seconds for follow-ups
Bangladesh Impact	20x increase in patients treated, 3x improvement in BP control

Simple | Publications and talks

Publication	Journal / Blog
Research Paper	Nature, Journal of Human Hypertension, 2024
Systems Thinking Approach	Simple.org Blog, 2024
Offline-first Apps	Simple.org Blog, 2020
Engineering to Embrace Change	Fragments Conference, 2017

Simple | Key Stakeholders

Key Stakeholders	Who they are
Tom Frieden	Former CDC Director, President of Resolve to Save Lives
Daniel Burka	Director of Product & Design (former Partner at Google Ventures, Digg)
Dr. Prabhdeep	Head of Epidemiology at ICMR for NCD
Dr. Reena Gupta	Clinical researcher at Johns Hopkins for treatment protocols
ICMR	Indian Council of Medical Research, Infrastructure and compliance
HISP	For DHIS2 national health system coordination
CVHOs	Chief District Health Officers from multiple Indian states

Simple | Technical Deep Dive

Offline-First Architecture for Rural Healthcare

- Built fully offline-first sync APIs enabling the app to work with zero connectivity
- Designed a phone-masking system using DTMF tones to protect nurse privacy in low-network areas
- Built a rapid-prototyping experiments app in ClojureScript + React Native to test UX with nurses in the field

Healthcare Interoperability & Standards

- Designed FHIR-compliant Import API with OAuth for external orgs (Bangladesh partners, OpenSRP in Sri Lanka)
- Strategic DHIS2 integration: disaggregated exports, tracker-to-analytics pipelines, facility-level dashboards
- Built DHIS2 Data Verifier: reduced data audit time from 1 day to 10 minutes
- Led CPHC migrations for Indian state health programs in coordination with ICMR

Infrastructure & Performance

- AWS migration completed urgently at 1/5th the original budget
- Reduced infrastructure costs by 5x while maintaining reliability
- Built health dashboards using materialized views and Russian Doll caching for sub-second queries across millions of records
- Established monitoring and observability practices (Prometheus/Grafana)

Simple | Tech Stack

Simple's source code is open source: github.com/simpledotorg/simple-server

Layer	Technologies
Backend	Ruby on Rails, PostgreSQL, Sidekiq
Mobile	Kotlin/Android, ClojureScript React Native
Infrastructure	Kubernetes, Ansible, AWS
Monitoring	Prometheus, Grafana

Consolidate Health

AI-Powered Healthcare Interoperability in the US

Consolidate Health | Overview

Consolidate Health is a US-based healthtech company focused on inter-EHR (Electronic Health Record) data consolidation. The product is a single API that gives application developers and healthcare platforms real-time access to structured, normalized clinical data from across the U.S. healthcare system.

They aggregate patient data from multiple EHR systems using the FHIR R4 specification to provide longitudinal health records once a patient authenticates and consents.

We have been working with Consolidate Health since February 2023.

Consolidate Health | What we built

Provider and Patient Applications

- Built two core applications: a Provider App for healthcare providers to view consolidated patient data, and a Patient App for patients to access and share their health records
- Built patient data sharing workflows enabling request-based data exchange between providers and patients

EHR Integrations

- Integrated with major US EHR systems including Athena, Modmed, iSalus, and Medent
- Enabling seamless data exchange across healthcare providers

FHIR-Compliant Data Pipelines

- Implemented FHIR R4 compliant data pipelines using Clojure's core.async for efficient healthcare data processing and synchronization
- Implemented FHIR endpoint search with multi-faceted and agentic search capabilities

Consolidate Health | What we built

AI-Powered Features

- Developed a RAG pipeline for patient summarization
- Agentic AI search for patients
- Optimized costs from $70/patient to $2/patient — a 35x cost reduction

HIPAA-Compliant Infrastructure

- Ensured HIPAA compliance on AWS infrastructure
- Resolved security vulnerabilities and established observability with Datadog

Consolidate Health | Tech Stack

Layer	Technologies
Backend	Clojure
Frontend	ClojureScript, Re-frame, Pushy
Infrastructure	AWS (HIPAA-compliant)
Standards	FHIR R4, HL7
Monitoring	Datadog
AI/ML	RAG pipelines, LLM integration

Staples

Enterprise-Scale Data & ML Systems

Staples | Overview

Staples Inc. is one of the largest American office retail companies. We worked with StaplesLabs/SparX, their innovation arm, building recommendations through email and search, providing shipping estimates, and an experimentation platform amongst other products.

We worked with Staples from 2013 to 2017 on multiple products.

Products We Built:

- Experimentation Platform: A/B testing and multivariate experimentation at scale
- Email Recommendation System: ML powered product recommendations for email marketing
- Web Crawler/Scraper: Competitive intelligence and pricing insights
- ETL & Data Warehousing: Analytics infrastructure for business insights

Staples | Technical Highlights

Experimentation Platform

- Engineered an experimentation platform incorporating research from white papers by Microsoft, Netflix, and Google
- Built a highly available system with automatic fail-overs
- Built a simulation testing harness using Causatum, Simulant, and Datomic

Machine Learning Runtimes

- Built and maintained ML runtimes for recommendation engines
- Email marketing personalization for existing customers

Data Engineering

- Orchestrated a web crawler and scraper handling unstructured datasets
- Provided insights into categorization and pricing on a daily basis

Staples | Conference Talks

Conference	Talk
RootConf 2016	On the Building of a Postgres Cluster
FnConf 2015	Building an Experimentation Platform
PgConf India 2017	A Postgres Orchestra

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